Aurora: Data and Algorithms

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Phase 1 needs seven streams from the inverter vendors' APIs and four outside inputs, and eight methods turn them into peak alerts, schedules and savings reports. One thing decides how much Aurora can do: a grid-point meter. Without one, Aurora can optimize solar and battery use but cannot see plant load or kVA demand.

Data to pull

The vendor APIs give inverter-side data. The poll interval is set by each vendor's rate limits, so plan for minutes, not seconds.

Stream	Fields to pull	Source	Cadence	Feeds
Plant and device registry	plant ID, timezone, installed kWp, inverter models, battery rated capacity	Vendor API	Daily	One common schema across vendors
Inverter real-time	AC active and reactive power, power factor, phase voltage and current, frequency, DC voltage and current per string, temperature, status	Vendor API	Vendor rate limit	Solar forecast, fault detection
Energy counters	hourly, daily and monthly yield (kWh); on-grid and self-use energy	Vendor API	Hourly and daily	Cross-check of integrated power, billing check

Stream	Fields to pull	Source	Cadence	Feeds
Battery	state of charge, charge and discharge power, voltage, current, temperature, working mode, SOC limits, rated capacity	Vendor API	Vendor rate limit	Dispatch, battery health
Grid meter (only where fitted)	import and export power, energy in and out, phase voltage and current, power factor	Vendor API	Vendor rate limit	Plant load, kVA demand, power factor
Irradiance sensor (only where fitted)	irradiance, module temperature	Vendor API	Vendor rate limit	Performance ratio
Alarms and events	fault code, device, start and end time	Vendor API	Every poll	Reliability insights
Tariff schedule	CI tariff category, supply voltage, energy rates and time-of-use windows, kVA demand charge, fixed charge, pass-through charges	KPLC and EPRA schedules	Entered per site; recheck at each tariff review	Cost model
Utility bills	billed kWh by period, maximum kVA, power factor, amount charged	Customer upload	Monthly	Billing check
Weather forecast	irradiance and cloud cover	Weather service	Hourly	Solar forecast
Plant schedule	shifts, shutdown days, flexible loads and their limits	Plant manager	At onboarding	Load-shifting constraints

What the first two vendors expose

- **Huawei FusionSolar:** the northbound API covers real-time and historical device data (string inverter, grid meter, battery), plant KPIs by hour, day, month and year, and alarms. It also accepts commands: battery working mode (maximum self-consumption, time-of-use, third-party dispatch), battery SOC limits and maximum power, and inverter active-power limits. Commands run as asynchronous tasks and must not be repeated before the previous task finishes. The reference documents flow control per API account.
- **Victron VRM:** the VRM API v2, as wrapped by an open-source MCP server, offers time-series statistics (live feed, consumption, solar yield, forecast), battery summaries with state of charge, PV inverter status, alarms and diagnostics. Access uses a personal token, and the wrapper notes a 200-request rolling limit. This comes from the wrapper, not Victron's own documentation.
- **Not yet checked:** other brands, for example Sungrow, Solis and Growatt. Confirm each one's API and rate limits before promising support.

Algorithms

Each method has one job and one output. Start with the simplest version that works, and add complexity only when it beats that version on held-out data.

Method	How it works	Output
1. Normalize and clean	Map each vendor's fields to one schema. Convert units, which differ by vendor and API version (Huawei changed its grid-meter active power unit to W in release 24.7.0). Align timestamps to Africa/Nairobi, flag stuck, negative and out-of-range values, and fill only short gaps.	One clean time series per site
2. Load reconstruction	With a grid meter, plant load = solar output + grid import − export ± battery (formula below). Without one, load is not observable, so report solar and battery only.	Plant load in kW

Method	How it works	Output
3. Forecasting	Baseline: the same hour last week. Move to gradient-boosted trees on lags, hour, weekday, shift calendar and forecast irradiance only if they beat the baseline on held-out weeks. Solar: clear-sky output scaled by forecast cloud cover and corrected on the plant's own history. Predict ranges, not only the mean, because peaks drive the cost.	24-hour load and solar forecasts with ranges
4. Peak management	The kVA demand charge follows the highest demand interval of the billing period, so only the top peaks matter. Alert when the upper forecast range passes the month-to-date peak; below that level, shaving saves nothing until the next bill. Actions: discharge the battery or shift a flexible load.	Peak alerts with lead time
5. Tariff-aware dispatch	Receding-horizon optimization: each cycle, solve a linear program over the next 24 hours (model below) using the forecasts, tariff windows, battery limits and flexible-load windows. Send only the first step, then re-solve. An open-source solver such as HiGHS is enough.	Battery schedule and load-shift plan
6. Power factor	Compute real power over apparent power per interval. Track the monthly average and the time spent below the tariff threshold. Size correction from the reactive power to remove (formula below).	Penalty exposure and capacitor size estimate

Method	How it works	Output
7. Fault detection	Compare each inverter and string with its expected output and with its siblings. Flag drops in performance ratio, string imbalance and repeated alarms, ranked by lost kWh.	Ranked fault list
8. Billing check and savings baseline	Recompute the bill from readings and the tariff, and flag gaps above a per-customer tolerance. Savings = baseline cost − actual cost, where the baseline is a weather- and production-adjusted model of what the bill would have been (the whole-facility approach of IPMVP).	Verified savings for reports and pay-on-savings pricing

Models behind methods 2, 5 and 6

Plant load, with grid import and battery discharge counted positive:

```
P_{load} = P_{pv} + P_{grid} + P_{batt}
```

Dispatch: minimize energy cost plus the demand charge over the horizon, where c_t is the energy price in interval t, d is the demand charge per kVA, and E is battery capacity. A power balance and the flexible-load windows are added as further constraints.

```
\min_{p,\,P_{peak}} \; \sum_t c_t \, p^{grid}_t \, \Delta t + d \, P_{peak}
\quad \text{s.t.} \quad p^{grid}_t \le P_{peak}, \quad
SOC_{t+1} = SOC_t + \frac{\eta_c \, p^{ch}_t - p^{dis}_t / \eta_d}{E} \,
\Delta t, \quad
SOC_{min} \le SOC_t \le SOC_{max}
```

Model in kVA when the meter reports apparent power; otherwise divide kW by the measured power factor.

Power factor correction, with P the real power, φ_1 the present angle and φ_2 the target angle:

```
Q_c = P \, (\tan\phi_1 - \tan\phi_2)
```

Insights delivery

Every insight carries five things: what to do, why, the KSh at stake, by when, and how sure Aurora is. Insights are ranked by expected KSh saved times confidence, so the plant manager sees the biggest one first.

Channel	Content	Trigger
Peak alert	forecast peak, its time, the suggested action	Only when the forecast passes the threshold and there is time to act; one alert per event
Daily digest	yesterday's peak, cost and savings, plus three ranked actions	Every morning
Monthly finance report	recomputed bill against the billed amount, demand charge, power factor, verified savings	Each billing date
Dashboard	live power against the peak threshold, battery state of charge, current tariff window	Always on

Send alerts through the channels the plant manager already uses, such as SMS, WhatsApp and email.

Advice becomes control in three stages:

1. **Advise only:** Aurora recommends, people act.
2. **Battery dispatch with manual override:** Aurora sends commands through the vendor API and the plant can cancel them.
3. **Automatic dispatch:** Aurora also returns the battery to the vendor's default mode if it loses connection or a command fails.

Because vendor commands run as slow asynchronous tasks, Aurora acts as a scheduler that plans ahead, not as a fast controller.

Build order and open questions

1. Confirm, for the pilot plant, the inverter brand, whether a grid meter is fitted, and who can create the API account.
2. Pull the longest history the vendor allows and build the baselines and forecasts from it.
3. Run advise-only through one full billing cycle and compare Aurora's numbers with the bill.

4. Enable battery dispatch only once forecast error is acceptable to the customer.

Open questions, with what we found

Power factor terms and the Victron rate limit are settled. The demand interval, the Huawei quota and the brand mix are not.

1. **Demand interval and power factor terms.** The tariff schedule defines demand as the maximum demand drawn in a billing period but does not say over what interval it is averaged. Make the interval a per-site setting, confirm it from the customer's Kenya Power meter data, and design alerts to work at 15 and 30 minutes until then. The power factor terms are clear and are in the table below.
2. **Inverter brands.** No public Kenyan breakdown found. Huawei is active in Kenyan commercial and industrial solar, with a C&I inverter series launched in Nairobi in August 2024 (Xinhua report), so build that integration first. Settle the rest with a short survey of 10 to 20 target plants through the IEEE PES and SIGHT network and its installers: inverter brand, whether a grid meter is fitted, and whether an API account exists.
3. **Rate limits.** Victron's VRM API allows a rolling window of 200 requests, with one request freed every 0.33 seconds (about 3 per second), and returns a Retry-After header on a 429 (Victron community, quoting the API docs). One user there was still blocked while polling every 15 minutes, so poll from one fixed server and back off on a 429. For Huawei, the quotas sit in the flow-control chapter of the northbound reference. Third-party tools report that real-time data refreshes about every 5 minutes and that faster polling returns error 407; I did not check this against Huawei's own text, so confirm the per-account quota before promising a poll interval.

Kenya Power rules that shape Aurora

From the Kenya Gazette tariff schedule, whose latest rates take effect on 1 July 2025.

Rule	What the schedule says	Effect on Aurora
Demand charge	Per kVA on the maximum demand in each billing period: CI1 (415 V) KSh 1,100; CI2 (11 kV) KSh 700; CI3 (33 kV) KSh 370; CI4 and CI5 KSh 300; CI6 and CI7 KSh 200	One shaved kVA saves up to that rate, once per bill

Rule	What the schedule says	Effect on Aurora
Off-peak windows	Weekdays 00:00 to 06:00 and 22:00 to 24:00; Saturdays and holidays 00:00 to 08:00 and 14:00 to 24:00; Sundays listed as 00:00 to 00:00, which reads as the whole day	Load-shifting windows in the dispatch model
Off-peak discount threshold	The discounted rate applies only to units above a monthly threshold: the last six months' average plus 6% (the first three months' average for new customers)	Shifting load earns the discount only above the threshold, so the model needs each site's threshold
Extra off-peak discount	5% off the off-peak rate for customers at 100% production capacity, once Kenya Power confirms it	Record as a per-site flag
Power factor	Below 0.90, Kenya Power may give 30 days' notice. If it is not fixed, 2% is added to the energy and kVA demand payment for each complete 1% below 0.90, until power factor is back to 0.90. Power factor is demand kW divided by demand kVA, measured by Kenya Power's own equipment	Target at least 0.90. At 0.85 the surcharge is 10% of those two charges, reading 1% as one point of power factor
Charges outside the surcharge	VAT, fuel energy cost, foreign-exchange adjustment, REP and EPRA levies are excluded from the surcharge base	The billing check must keep these lines separate

Check before quoting savings: a news report says EPRA amended the schedule on 18 September 2026. I did not check whether that changed the commercial and industrial terms above.

Sources

- Huawei SmartPVMS 25.2.0 Northbound API Reference
- Victron VRM MCP server README

- Kenya Gazette Notice 3899: Schedule of Tariffs 2023, with rates to 1 July 2025
- Victron community thread quoting the VRM rate-limit documentation
- Xinhua report on Huawei’s 2024 Nairobi C&I inverter launch